Exporting Map

Macro Data

- Economic Survey 2024
  - % of GDP — 18%
  - Growth rate over last 5 years — 4.2%
  - % of total exports
    - 11.7%
    - $53 Bn
  - % of workforce — 45%

Issues related to direct and indirect farm subsidies and minimum support prices

- Subsidy
  - Data
    - Total subsidy expenditure (FY 2025–26): ₹4.26 lakh crore
    - Increased from ₹4.03 lakh crore (FY 2023–24)
    - Food subsidy — 2.03 lakh crore
    - Fertiliser subsidy
      - 1.68 lakh crore
      - 35% dependency — met through imports
      - $9 Bn imports
    - Power Subsidy & Agriculture
  - Direct
    - Subsidies directly transferred to farmers, usually in cash form with Aim to reduce leakages and improve targeting
    - examples
      - PM-KISAN — DBT of 6000 to farmers per year
      - Rythu Bandhu Scheme — India's first direct cash transfer scheme providing financial support to farmers for farm inputs like seeds, fertilizers, and labor, giving them ₹5,000 per acre per season (Kharif & Rabi), aiming to break debt cycles and boost productivity without being a loan.
      - PAHAL (LPG DBT) — an Indian government initiative for the Direct Benefit Transfer (DBTL) of subsidies for Liquefied Petroleum Gas (LPG). The primary goal is to ensure transparency and efficiency by transferring the subsidy amount directly into the bank accounts of eligible consumers.
      - Farm loan waivers
    - Benefits
      - Enhances purchasing power of farmers
      - Provides freedom of choice in expenditure
      - Improves resource-use efficiency — Farmers buy inputs (e.g., fertilisers) at market price → avoid overuse
      - Cash transfers act as working capital for agriculture
      - Lowers government administrative costs — Reduced storage, transport, and procurement burden
    - Challenges to direct farm subsidies
      - Inadequate financial inclusion of farmers
      - Potential inflationary impact
      - Errors in identification of beneficiaries
      - Limited access to markets restricts effective utilisation
  - Indirect farm subsidies
    - Subsidies where input prices are kept below market prices — Provided through price controls or concessions
    - Types
      - Tax concessions
      - Insurance support — MSP
      - Low-interest credit — KCC credit cards
      - Depreciation write-offs
      - Rent and input rebates
    - Data — India spends around 2% of GDP on indirect subsidies
    - examples — electricity subsidy, fertiliser subsidy
    - MSP
      - Intro — MSP is the minimum price that the government considers remunerative for agricultural production. In event of fall of prices in notified agricultural commodities, the government provides price support by procuring the farm produce at MSP.
      - Data
        - Covered — 24
          - torai — rapeseed and mustard
          - Dehusked Coconut — Cpora
        - Post harvest losses — 15-25% — ICAR
        - commodities — 23
        - Fair and Renumerative Price for sugarcane
        - recommended by — Commssion for agri cost and prices (CACP)
        - fixed by — Cbinet Committee on economic affairs
      - Benefits
        - ensures food secutity and affordability — Public Distribution System (PDS), supported by procurement at MSP, has been central to managing food economy — contain the rise in food prices and supply
        - It incentivizes cultivators to adopt modern technology and raise productivity in line with emerging demand patterns — during green revolution productivity enhanced due to combination of MSP and tech advancements
        - Indirect subsidies help stabilize markets for basic foodgrains. — MSP protect farmers from sharp falls in prices, particularly when international prices drop or market volatility occurs
        - Protection from fluctuation in global markets and food security is achieved
        - Alleviation of Rural poverty prevents rural to urban migration for jobs
      - Challenges
        - Limited Reach and Inequity — Data from the 70th round of NSSO (2012-13) revealed that only 6% of total farmers gained from selling wheat and paddy directly to any procurement agency
        - Regional Disparity: — Procurement operations are highly concentrated in states like Punjab, Haryana, Andhra Pradesh, and Madhya Pradesh
        - Fiscal Burden
          - High Cost: Food and fertilizer subsidies constitute 87% of the total subsidy bill — 4L cr
          - subsidies can distort market to ultimately hurt the poor — Farm loan waiver not only increases the states' fiscal deficit and interest burden, but also limits their ability to invest in productive capital in agriculture.
        - Environmental and Resource Inefficiency
          - Imbalanced Fertilizer Use:
            - The heavy subsidy on urea compared to Phosphatic and Potassic fertilizers has created a highly imbalanced use of Nitrogen, Phosphorus, and Potassium (NPK)
            - Subsidies on chemical fertilizers make them cheaper than organic options, creating an uneven playing field — This discourages the use of organic manures and affects soil health
          - Water Depletion: — Policies such as free power for agriculture, combined with the dominance of water-guzzling crops like paddy (incentivized by MSP), have caused indiscriminate use of groundwater. — This has turned large tracts of the Indo-Gangetic Plains into "dark zones" due to over-exploitation
        - Distortion of Cropping Patterns — skewed incentive structure where farmers prefer wheat and rice crops over pulses and oilseeds, even though the country is deficit in the latter
        - Adverse effect on sugar mills — when State administrtive prices in set over and above FRP
        - Operational Inefficiencies
          - Buffer stocks with the Food Corporation of India (FCI) have often been more than double the norms costing the nation thousands of crores without serving a worthwhile purpose.
          - Storage Losses — Due to inadequate covered storage capacity, substantial quantities of grains are stored in Cover and Plinth (CAP) storage, leading to deterioration — For instance, in Punjab, 78% of the damaged wheat stock was attributed to CAP storage
      - Reforms
        - Minimum Support Price (MSP) and Procurement
          - The Swaminathan Committee recommended that MSP should be at least 50% more than the weighted average cost of production
          - Diversification of Procurement:
            - government must provide better price support for pulses and oilseeds and
            - dovetail MSP policy with trade policy to prevent landed costs from falling below MSP
          - Decentralization:
            - handing over procurement operations of wheat and rice to states with sufficient experience (pujab and haryna)
            - shifting the Food Corporation of India's (FCI) focus to eastern states (Bihar, UP, West Bengal) where farmers suffer from distress sales
            - Private Sector Participation: — The DFI Committee recommended permitting private stockists to procure at MSP with valid incentives
        - direct cash subsidy of about ₹7,000/hectare to farmers — Shanta Kumar
    - Reforms
      - Agricultural Marketing and Logistics Reforms
        - The DFI Committee recommended a new market architecture comprising 22,000 Gramin Agricultural Markets (GrAMs) at the village level (upgrading existing haats), linked to competitive wholesale markets and export markets
        - Single Indian Market — State APMC Acts should be amended to promote grading, branding, and the development of domestic and international markets, moving towards a Single Indian Market
      - Institutional and Structural Reforms
        - Restructuring FCI — FCI be restructured into an agency for innovations in food grain management,
          - handle bulk supply chains
          - end to end computerisation
        - Land Reforms
          - It also suggested establishing a National Land Use Advisory Service
          - distributing ceiling-surplus and waste lands and preventing the diversion of prime agricultural land for non-agricultural corporate purposes
        - States should adopt the Model Contract Farming and Services Act to facilitate high-tech cultivation and better market linkages
        - Contract Farming: States should adopt the Model Contract Farming and Services Act to facilitate high-tech cultivation and better market linkages
        - agriculture shall be placed in concurrent list of constitutions
      - NITI Ayog recommendations — Sustainability and Organic Inputs (Gaushalas)
        - Parity in Assistance: There should be parity in support for chemical fertilizers vis-à-vis organic/cow-dung-based fertilizers to create a level playing field
        - Public sector fertilizer agencies (like IFFCO, KRIBHCO) should be mandated to market standardized organic and bio-fertilizers produced by Gaushalas
        - The Fertilizer Control Order (FCO) should be expanded to include products like Panchagavya, Jivamrit, and Ghanjivamrut to facilitate their commercial production and quality control
        - Promote the production of Compressed Bio Gas (CBG) using agri-biomass and cattle waste, similar to the GOBARdhan scheme
      - Crowding Out Private Sector:
        - High statutory levies and commissions (Mandi Tax) imposed by some states (e.g., up to 14.5% in Punjab) on procurement crowd out the private sector
        - bonuses declared by states over the MSP distort the market, further discouraging private participation
  - WTO Issues Related to India's Farm Subsidies
    - Amber Box: — Trade-distorting subsidies (e.g., MSP)
      - MSP above Deminimis level (10% of agri output value)
      - Stockholding and MSP — Procurement at MSP alleged to distort global prices
    - Green Box: — Minimally trade-distorting (e.g., R&D, infrastructure) — DBT (Ryathu bandhu scheme)
  - Other issues — Failure of states to ensure rational use of subsidised water and electricity

Main Topic 15

Public Distribution System- objectives, functioning, limitations, revamping;

- PDS
  - Intro
  - Data
    - Beneficiaries
      - 80Cr +
      - 57% of pupulation
    - items — rice, wheat, sugar, kerosene
  - NFSA
    - covering up to 75% of the rural and 50% of the urban population
    - It categorizes beneficiaries into Antyodaya Anna Yojana (AAY) (poorest of the poor) and Priority Households (PHH)
    - Under the Pradhan Mantri Garib Kalyan Anna Yojana (PMGKAY), the government provides 5 kg of free food grains per person per month to priority households and 35 kg per family per month to AAY households. This scheme was extended to provide free grains effective from January 2023
  - Issues
    - leakage
      - 69000 Cr+
        - ICRIER study
        - 28% of food grains
      - 40-50% — shanta kumar committe
    - Nutritional deficiency due to major empahsis on rice and wheat
    - Fiscal impact — ~2L Cr
    - tageting errors — old data of 2011
    - storage losses — 10% of annual spoilage
    - corruption at FRP — underweighing, overchargins
  - govt stance
    - Tech reforms
      - Digitization of ration cards and beneficiary databases to remove duplicates.
      - Mandatory Aadhaar linkage to prevent ghost beneficiaries.
      - 95% of Fair Price Shops are equipped with PoS machines for biometric authentication — to reduce the leakage from
        - 46% — 2011-12
        - 28% — 2022-23 — HCES, 2022-23
    - Ration Card Portability: One Nation One Ration Card (ONORC): Enables access to PDS benefits nationwide, aiding migrant workers.
    - Doorstep delivery of foodgrains to FPSs in many states.
    - Nutritional Initiatives: Distribution of fortified rice under POSHAN Abhiyan to combat anemia and malnutrition.
    - Infrastructure Improvement: Construction of modern silos, cold storage, and warehouses
  - Reforms
    - Reduce coverage from 67% to 40%, focusing on BPL families.
      - Recent policy briefs suggest providing free food only to the bottom 15% (extreme poverty), while offering grains at 50% of MSP to the rest of the beneficiaries
      - Increase grain allocation from 5kg to 7kg per person
      - Shanta kumar committee
    - Fiscal Reforms — Direct Cash Transfers: — Reforms suggest a gradual shift towards Direct Benefit Transfer (DBT) or cash transfers for food security, starting with large cities and surplus states — curbing leakage, reduce administrative cost, empower beneficiary to purchase grains of their choice
    - Fair Price Shops (FPS) could be transformed into nutrition hubs, potentially linked with a food coupon system to ensure access to diverse and nutritious food
    - A transparent liquidation policy is needed for buffer stocks to prevent holding excessive stocks that cost the nation heavily without serving a purpose
    - Provide 6 months' ration to beneficiaries after procurement season. Reduce storage costs and consumer hassles.
    - storage. Containerize grain movement to minimize losses.

Main Topic 16

issues of buffer stocks and food security;

- Buffer stocks
  - Data
    - furtis and veggies transport via roadways — 97%
    - Requirement — 70 mT
    - Availability — 35 mT
    - Shanta Kumar Committe Report
  - Issues
    - Excessive Stockholding: A major challenge is that buffer stocks held by the Food Corporation of India (FCI) have often been more than double the prescribed norms.
    - lack of norms for maxmimum level of stocks to be maintained
    - Storage Capacity Constraints: — FCI's owned covered capacity is only 32% of the total requirement — Consequently, substantial quantities of grain are stored in Cover and Plinth (CAP) (open storage) or hired godowns. In Punjab alone, nearly 78% of damaged wheat stock was attributed to CAP storage
    - Due to inadequate scientific storage and prolonged storage of grain, significant quantities are damaged — Between 2017-18 and 2021-22, 13,349 MT of food grains valued at ₹35.57 crore were damaged — CAG
  - Govt steps
    - Private Entrepreneur Guarantee (PEG) Scheme — private parties construct godowns which FCI hires with a guarantee of usage — As of March 2022, 14.5 MMT capacity was completed under this scheme
    - Silo Construction: — action plan to construct 100 LMT of steel silos in Public-Private Partnership (PPP) mode. — progress has been slow — with only 12.25 LMT constructed by FCI against a target of 29 LMT under the policy
    - Open Market Sale Scheme (OMSS) — government utilizes OMSS to offload excess stock in the domestic market to control prices and liquidate surplus
    - Price Stabilization Fund: A Price Stabilization Fund is in place (with a budget of ₹4,019 crore for 2025-26) to manage price volatility
  - Reforms required
    - A. Structural and Policy Reforms
      - Transparent Liquidation Policy: The Shanta Kumar Committee (HLC) and CAG recommend a pro-active liquidation policy that automatically kicks in when stocks exceed buffer norms. This involves selling surplus in the open market (OMSS) or export markets to reduce costs
      - Revisiting Buffer Norms: The norms need to be revised to clearly delineate between operational stocks and strategic reserves.
      - A maximum stock limit should be prescribed to prevent unlimited accumulation
      - Decentralized Procurement: The HLC recommended handing over procurement of wheat and rice to states that have gained sufficient experience (e.g., Punjab, Haryana, MP) and focusing FCI's attention on deficit states to prevent distress sales
    - B. Infrastructure and Storage Reforms
      - Phase Out CAP Storage: Covered and Plinth (CAP) storage should be gradually phased out, with no grain remaining in CAP for more than 3 months. It should be replaced by silos and conventional godowns.
      - • Shift to Silos and Bulk Handling: There should be a massive shift from bagged storage to bulk handling using steel silos. This promotes a "machine back" rather than "human back" approach, reduces losses, and improves shelf life.
      - • Private Sector Participation: FCI should outsource stocking operations to the Central Warehousing Corporation (CWC), State Warehousing Corporations (SWC), and the private sector on a competitive bidding basis to reduce costs
      - Agri-Logistics: The DFI Committee recommended promoting integrated agri-logistics (pack-houses, reefer transport) to reduce post-harvest losses and facilitate the "Fork to Farm" approach
    - C. Market Reofrms
      - Negotiable Warehouse Receipts (NWRs): The NWR system should be scaled up. Farmers can deposit produce in registered warehouses and get bank advances (say 80% of MSP) against it. This reduces the need for physical procurement by the government and allows farmers to sell when prices are better.
      - • Direct Benefit Transfer (DBT): A gradual shift to cash transfers in the Public Distribution System (PDS) is recommended for large cities and surplus states. This would reduce the physical handling of grain, plug leakages (saving an estimated ₹30,000 crore annually), and reduce the pressure on buffer stocks.
      - • Movement Optimization: FCI should strictly follow Linear Programming to minimize the cost of transporting grains. Currently, deviations from optimal routes result in higher freight costs. Movement should be gradually containerized to reduce transit losses
- Food Security
  - Definition
    - when all people at all times
    - have physical and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life.
    - FAO
  - Intro — While India has achieved self-sufficiency in grain production, food security challenges have shifted from availability to accessibility, nutritional quality, and system efficiency.
  - Issues
    - Calorie vs. Nutrition: The current system is heavily cereal-centric (rice and wheat), which provides calories but lacks essential micronutrients. This has led to "hidden hunger" or micronutrient deficiency — 35.5% of children under five are stunted, 19.3% are wasted, and 32.1% are underweight
    - Dietary Imbalance: Household consumption surveys (2022-23) reveal that the share of spending on nutritious items like pulses and vegetables in total food expenditure has declined, while spending on processed foods has increased
    - leakage — 28% of PDS — 69000 cr
    - regional disparity — Leakages are significantly higher in certain states, such as Arunachal Pradesh (63.2%) and Nagaland (60.4%), compared to states with better tracking
    - Coverage Errors: There are issues of both exclusion (needy excluded) and inclusion (non-eligible included) errors in the beneficiary lists
  - Govt steps
    - govt steps under PDS sections — nfsa, pmgkay,digitisation, epos, one nation one ration card
    - Poshan Abhiyan: Initiatives like Saksham Anganwadi and POSHAN 2.0 have been allocated ₹21,960 crore (2025-26 BE) to improve nutritional outcomes for children and mothers
    - Bio-fortification: The government is promoting bio-fortified varieties of crops (e.g., zinc-rich rice, iron-rich pearl millet) to address micronutrient deficiencies
  - reforms
    - PDS related — dbt , reduction in coverage, updated beneficiary list, diversification , food coupan and stamps at FPS
    - storage reforms — silo, containsersiation, warehouse, farm to fork infra, etc.
    - shift towards millets — Intl. year of millets 2018
    - Life-Cycle Approach: Reorganizing nutrition support programmes on a life-cycle basis, targeting specific nutritional needs from infancy to old age
    - Community Food Banks: Promoting the establishment of Community Food and Water Banks operated by Women Self-help Groups (SHGs) to ensure local food security
    - National Food Guarantee Act: The Swaminathan Committee suggested formulating a National Food Guarantee Act that combines the features of food for work and employment guarantee programmes

Main Topic 17

economics of animal-rearing.

- Livestock
  - Status
    - This accounts for 30.23% of the Agricultural & Allied Sector GVA and 5.50% of the total national GVA,. The sector grew at a Compound Annual Growth Rate (CAGR) of 7.67% between 2014-15 and 2021-22 at constant prices
    - Milk — Milk: India is the largest milk producer in the world, contributing 25% to global production
    - Eggs: India ranks 2nd globally in egg production
    - India ranks 4th globally in meat production,
    - The sector provides livelihoods to approximately 8.5 crore rural households — DAHD annual report 2024-25
  - Importance
    - boost exports positively impacting BoP — leather, meat, wood, etc.
    - empolyement generation in rural households
    - integerated farming would lead to reduction of chemical fertilisers reducing import bill and improving soil health
    - combating hidden hunger
    - Risk mitigation by providing a sense of security and confidence to landless, small, and marginal farmers
    - Biogas production and sustainable agricultural development
  - Issues
    - Feed and Fodder Deficit: There is a critical shortage of feed. The country faces a net shortfall of 35.6% in green fodder, — NITI aayog
    - Low Productivity: Despite high total production, per-animal productivity remains a challenge. For instance, the average milk yield for indigenous cows is significantly lower (4.20 kg/day) compared to exotic cows (9.82 kg/day) — DAHD
    - Disease Outbreaks: Diseases like Foot and Mouth Disease (FMD) and Brucellosis cause massive economic losses, estimated at around ₹24,000 crore annually for FMD alone — DAHD
    - Stray Cattle and Unproductive Animals: — male cattle and unproductive cows are often abandoned, leading to a stray cattle population estimated at 53 lakhs,. This causes crop damage and road accidents
    - Artificial Insemination (AI) coverage is limited to 33% of breedable bovines, with the remaining 70% covered by scrub bulls of unknown genetic merit
    - Viability of Gaushalas: Most Gaushalas (cow shelters) depend on donations and lack sustainable revenue models. Operating expenses often exceed income, making them financially vulnerable
  - Govt stps
    - Rashtriya Gokul Mission (RGM): Launched to develop and conserve indigenous bovine breeds. It focuses on enhancing productivity through technologies like IVF and Sex Sorted Seme — susbidies for establishment of Breed Multiplication Farms and for IVF pregnancies
    - National Livestock Mission (NLM)
      - Realigned to focus on entrepreneurship development in poultry, sheep, goat, and piggery
      - It provides capital subsidies (up to 50%) for establishing breeding farms to individuals, FPOs, and SHGs
    - Health
      - National Animal Disease Control Programme (NADCP): A central sector scheme to control FMD and Brucellosis through mass vaccination (e.g., FMD vaccination by 2025 and eradication by 2030)
      - Pashu sanjeevni — animal welness programme encompassing provision of animal health card along with UID identifications
      - National Aritificail Insemination Programme — to prevent spread of disease and sex-sorted breeding
    - Animal Husbandry Infrastructure Development Fund (AHIDF) — to incentivize investments in dairy/meat processing, animal feed plants, and waste-to-wealth management. It offers 3% interest subvention on loans
    - Kisan Credit Card (KCC): The facility of KCC has been extended to animal husbandry and fisheries farmers to meet their working capital needs
    - Digital Initiatives: Launch of the "Bharat Pashudhan" database and the 1962 mobile app to create a digital ecosystem for livestock management
    - Mobile Veterinary Units (MVUs): Financial support is provided to states for MVUs to deliver veterinary services at farmers' doorsteps
    - E pashu haat protal — connecting breeders and farmers regarding availability of bovine germplasm
  - reforms required
    - Promotion of Sex Sorted Semen: To address the issue of stray cattle and male calves, the widespread use of sex-sorted semen (producing female calves with >90% accuracy) is recommended.
    - "Cow Economy" and Waste to Wealth: — Bio-Fertilizers: Promoting the production of Phosphate Rich Organic Manure (PROM) and vermicompost from cow dung to make Gaushalas economically viable and reduce reliance on chemical fertilizers
    - Bio-Energy: Encouraging the setting up of Bio-CNG and biogas plants using cattle waste, linked with schemes like GOBARdhan and SATAT
    - Fodder Development: utilizing non-arable and wastelands for fodder production through agroforestry to bridge the supply deficit. Incentivizing fodder storage (silage/hay) entrepreneurs
    - Vetenary — One Health Approach: Strengthening the "One Health" framework to mitigate risks of zoonotic diseases (diseases transmitting from animals to humans) through integrated surveillance and "Standard Veterinary Treatment Guidelines"
    - Institutional Strengthening: Forming Village Producer Organisations (VPOs) for small ruminants and poultry to improve access to credit and markets,. Strengthening the cooperative credit structure for long-term lending to the sector
    - Export Focus: Improving compliance with food safety standards of Codex Alimentarius Commission to boost exports of dairy, meat, and poultry products, where India currently has untapped potential
    - NITI Ayog — bredding indegeneous cattles with exotic to enhance producitivity
- Aquaculture
  - Status
    - second-largest aquaculture producer and the 4th largest exporter of fish and fisheries
    - Structural Shift: There has been a significant shift from marine capture to inland aquaculture. Inland fisheries now constitute about 75% of total production, a major increase from 50% in 2000-01
    - 7.26% of the agricultural GVA
    - annual average growth rate of 6.30% over the last five years
    - Exports — USD 7.38 billion — FY 2023-24
    - The sector supports the livelihood of approximately 28 to 30 million people
  - Issues
    - Infrastructure Deficits:
      - insufficient cold chain connectivity from landing sites to retail markets, leading to post-harvest losses and hindering domestic demand in the hinterlands
      - The lack of strong value chains also hampers export competitiveness
      - lack of specifically fishing harbours, landing centers, cold chains, and distribution systems. — leads to post-harvest losses, poor processing, and value addition
    - Technological Gaps: — shortage of quality fish seeds, lack of mapping for fish-rich zones, and the use of obsolete fishing technology (bottom thrwaling)
    - Ecological Concerns:
      - Over-exploitation of marine resources and depletion of inland natural waters are major concerns
      - Pollution: Marine plastic litter is a growing threat
      - Marine capture fisheries face limited scope for expansion due to overcapacity in territorial waters, weak regulation, and prevalence of traditional fishing practices
    - Marketing and Financial Issues: — The share of high-value processed products is low (<15%) compared to global competitors like China (49%)
    - Inland and Aquaculture Challenges:
      - Resource Issues: Seasonal nature of operations, depleted stocks in natural waters
      - Input Constraints: Shortage of quality seed and feed, low productivity, and high input costs
      - Freshwater aquaculture is heavily dependent on Indian Major Carps (Catla, Rohu, Mrigal), lacking diversification
      - Export Quality Concerns: — "Salmonella" and veterinary drug residues are primary causes for export rejections in crustaceans
      - Sustainability Threats: The sector faces threats from marine plastic litter, pollution, and climate change affecting ecosystems
  - Govt steps
    - Pradhan Mantri Matsya Sampada Yojana (PMMSY): — address critical gaps in the value chain,
    - Infrastructure Development (FIDF): The Fisheries and Aquaculture Infrastructure Development Fund (FIDF) (fund size ~₹7,522 crore) provides concessional finance to state governments and private entities for developing fishing harbours, cold storages, and feed mills
    - Kisan Credit Card (KCC): Extended to fisheries farmers to meet working capital needs. The lending limit was recently increased to ₹5 lakh in the 2025-26 budget
    - Technology and Modernization:
      - Launch of the National Fisheries Digital Platform (NFDP) to create a registry of fish workers and enterprises
      - Vessel Communication: National Rollout Plan for Vessel Communication and Support System (VCSS) involving transponders for safety and communication
      - Disease Monitoring: Launch of the Report Fish Disease (RFD) App and the National Surveillance Programme for Aquatic Animal Diseases (NSPAAD)
      - Disease Monitoring: Launch of the Report Fish Disease (RFD) App and the National Surveillance Programme for Aquatic Animal Diseases (NSPAAD)
      - Implementation of the National Plan of Action on Sharks and engagement in the GloLitter partnership to reduce marine plastic
    - MGNREGS for developing fish ponds
  - Reforms required
    - Legal and Regulatory Framework:
      - Enactment of a comprehensive National Marine Fisheries (Regulation and Management) Bill to regulate the Exclusive Economic Zone (EEZ)
      - Adoption of a Model Leasing Policy by states to facilitate mariculture and secure tenure rights for small-scale fishers
    - Shift to Blue Economy and Deep Sea Fishing:
      - Encouraging traditional fishers to undertake deep-sea fishing for high-value resources like Tuna (especially in Andaman & Nicobar and Lakshadweep)
      - Designating areas for Mariculture Technology Parks and promoting seaweed cultivation
    - Value Chain and Marketing:
      - Cluster Approach: Developing integrated production and processing clusters to enable focused development of export-oriented species
      - Traceability: Implementing IT-enabled traceability and labeling systems from 'bait to plate' to ensure quality and meet international standards
      - Cold Chain: Massive expansion of cold chain infrastructure to reduce post-harvest losses from the current high levels to below 10%
    - Institutional Strengthening:
      - Promotion of Fish Farmer Producer Organizations (FFPOs) to enhance bargaining power and market access
      - Establishment of a National Fisheries Development Council for overall guidance and coordination between Center and States
      - Diversification: Moving beyond carps to culture exotic species and high-value native species (like Scampi, Tilapia, and Trout) in inland waters

Technology missions;

- To be done with cropping , irrigation, etc.

Land Reforms

- Data
  - Landless Households — [Socio Economic Caste Census 2011] — 30%
  - Negative
    - %person own less than 2 hectare [NSSO] — 90%
    - agri cencus 2018 farmer data
      - Landless — 7
      - Marginal — 75
      - Small — 10
  - Positive — Small and marginal owned area
    - 1961 — 20%
    - 2013 — 53%
  - In 1991-92, the bottom 50% of rural households owned only 3% of the total land, while the top 10% owned 54% — NCF
  - Fragmentation: The average size of operational holdings has declined significantly, from 2.67 hectares in 1960-61 to 1.34 hectares in 1990-91 — NCF
  - In 1991-92, 11.24% of rural households were strictly landless — NCF
  - Currently, small and marginal farmers own 86% of the land in India — Model Contract Farming and Services Act, 2018
- Introduction
  - The agrarian economy of India has historically been characterized by inefficiency, low productivity, and small units of production
  - Agrarian Reforms Committee (Kumarappa Committee) recommended in 1949 that there should be no place for intermediaries and that "land must belong to the tiller". It viewed land reforms as the "touchstone of democracy" essential for freedom from poverty
- Issues — The colonial land tenure systems—Zamindari, Mahalwari, and Ryotwari—created a disconnect between the cultivator and the land, leading to exploitation
- Reforms so far (MoRD)
  - Abolition of Intermediaries (1950-1972): Removing Zamindars and Jagirdars to bring cultivators in direct contact with the State — Intermediaries were abolished over 6.0 million hectares, and ownership rights were granted to 20 million tillers
  - Tenancy Reforms: Providing security of tenure, regulating rents (recommended at 20-25% of gross produce), and conferring ownership rights to tenants
    - Rights of 12.42 million tenants were protected over an area of 6.25 million hectares
    - successful in kerala and WB
    - Operation Barga: In West Bengal Operation Barga was launched in 1978 with the objective of achieving the registration of sharecroppers and providing them permanent occupancy and heritable rights and a crop division of 1:3 between landowner and sharecropper.
  - Land Ceiling Acts: Fixing statutory limits on land ownership and redistributing surplus land to the landless — Redistribution: An area of 2.12 million hectares of ceiling-surplus land was distributed to 5.67 million rural poor (51% of whom were SC/ST)
  - Consolidation of Holdings: Merging scattered landholdings into compact blocks to improve farming efficiency — About 65.34 million hectares of land was consolidated, notably in Punjab, Haryana, and Uttar Pradesh
  - Forest Rights Act, 2006: Recognized rights of forest-dwelling Scheduled Tribes and other traditional forest dwellers
  - Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (LARR Act): — Replaced the colonial 1894 Act to ensure humane, participatory processes and fair compensation for land acquisition
  - Recent steps
    - Model Agricultural Land Leasing Act, 2016: Proposed by NITI Aayog to legalize land leasing, ensuring security for land owners and access to credit for tenants
    - Model Land Titling Act: The central government has drafted a Model Land Titling Act, aiming to standardise procedures and facilitate a shift towards conclusive titling.
    - The Registration Bill 2025: Aimed at streamlining land transfers and ensuring that registration reflects actual ownership, not just transactions
    - DILRMP (Digital India Land Recored Modernisation Programme)
      - Digital India Land Records Modernization Programme (DILRMP): Aimed at digitizing land records
      - ULPIN / Bhu Aadhar — 14 digit alpha-numeric for each land parcel
      - GPS Based
      - integerated with land records
      - As of recent data, 94% of Rights of Record (RoR) have been computerized — 2024, MoRD
    - PM SVAMITAVA
      - 6.62 lakh villages in the country which will be eventually covered in this scheme.
      - clear ownership of property in 'Abadi' in habited area
      - It is a scheme for mapping the land parcels in rural inhabited areas using drone technology and Continuously Operating Reference Station (CORS) to provide property cards.This Scheme comes under the Ministry of Panchayati Raj.
- Factors Responsible
  - reforms succeeded (e.g., West Bengal's Operation Barga, Kerala),
    - Strong political commitment allied with peasant movements challenged the resistance of landed interests
    - Record of Rights: Successful states prioritized the updating of land records (e.g., recording sharecroppers in West Bengal)
    - Decentralization: Involvement of Panchayati Raj Institutions (PRIs) in identifying beneficiaries and managing land
    - Kisan Sabhas and Farmers Associations also helped farmers organize themselves and raise their demands.
    - Judicial backing and progressive interpretations of constitutional provisions aided in land reforms. Egland reform legislations through IX schedule.
  - Failure
    - Lack of Reliable Records: The absence of accurate land records made it difficult to identify surplus land or protected tenants
    - Legal Loopholes & Litigation: Large amounts of surplus land remained locked in court cases. Landowners used loopholes like Benami transactions to evade ceiling laws
    - Administrative Apathy: The revenue administration often lacked the vision and dynamism required for a holistic view of land management — NCF
    - Alienation of Tribal Land: There has been widespread dispossession of Dalits and Tribals from lands legally allotted to them
- Land Reforms and Agriculture Productivity
  - Debate
    - higher farm szie enhances productivity due to mechanisation, and economy of scale
    - small farm size increase productivity due to resource efficiency, high labor intensity and supervision
    - Productivity is linked to the size of holding; Indian productivity is much lower than global peers (e.g., paddy yield is 2929 kg/ha in India vs. 6321 kg/ha in China)
  - Reforms — Security of tenure encourages investment in land improvement. However, excessive fragmentation (minuscule holdings) hampers modern technology adoption To address this, contract farming and producer organizations (FPOs) are recommended to aggregate produce without dispossessing farmers of their land ownership
- Land Records
  - Issues
    - Records are often outdated and do not reflect ground realities (e.g., sharecroppers not recorded). — 60% of holdings still lack "conclusive titles," leading to endless litigation.
    - Discrepancy between revenue records and registration data — NCF
    - Slow progress in the survey/resurvey of land parcels
    - multiple claims — Complex inheritance and land transfer systems often result in overlapping claims and ownership ambiguity.
  - Land tilting
    - Current Status (Presumptive Titling)
      - India follows a system where land records are fiscal (for revenue) and presumptive.
      - Registration of deeds under the Registration Act (1908) records the transaction, not the title itself.
      - Ownership is proven through a chain of documents
    - Reforms Required (Conclusive Titling): There is a move toward Conclusive Titling, where the State guarantees the title. This requires a single, integrated agency for maintaining land records
      - Benefits
        - Reduced Litigation: More than 55% of civil disputes in India are land-related. Clear titles would drastically reduce this litigation
        - Access to Credit: Clear titles facilitate easier access to institutional credit (mortgages)
        - Greater Market Efficiency: Accurate and transparent land records can facilitate land transactions and promote a more efficient land market.
      - Challenges: The transition requires massive investment in surveying, updating records, and legal changes.
- Current Issues
  - Restrictive Tenancy Laws: Bans on leasing in many states prevent land from being used efficiently and leave tenants without legal protection — Kerala
  - Exclusion: Women, despite legal rights (Hindu Succession Amendment Act 2005), often lack actual control or titles (Joint Pattas) — women constitute only 14% of all operational landholders
  - Financial Exclusion: Tenant farmers cannot access formal credit (Kisan Credit Cards) due to lack of land titles
- way ahead
  - legal front
    - Legalize Land Leasing: States must adopt the Model Agricultural Land Leasing Act, 2016 to allow land owners to lease out land without fear of losing ownership and to enable tenants to access credit/insurance
    - Special Land Tribunals: Establish fast-track tribunals to resolve the 7.4 lakh acres currently locked in litigation (Economic Survey 2024-25 context).
    - Inclusive Definition of Farmer: Delink the definition of "farmer" from land ownership to include tenant farmers, sharecroppers, and women for accessing government benefits
  - Innovative
    - Land Pooling: Promote Farmer Producer Organizations (FPOs) and group farming to create economic scales of operation for smallholders
    - Contract Farming: Adopt the Model Contract Farming Act to link farmers with markets and technology while protecting land rights
  - Digitization: Complete digitization of land records on a "Mission Mode" including geo-tagging to reduce disputes
    - Digitizing Maps: Completing the 68% digitization of cadastral maps (DoLR 2024) to accurately identify encroached or surplus government land.
    - Administrative changes at the state level that streamline the collection and maintenance of land data, for example - Karnataka's BHOOMI Project.
  - Draft National Land Reforms Policy: The draft has five goals
    - Restore land unjustly taken from vulnerable communities such as Dalits (untouchables) and Tribals.
    - Protect the land of the Dalits and Tribals including the commons that they depend on going forward
    - Liberalize leasing laws.
    - Improve land rights of women.
    - Appropriate amendment to Forests Right Acts, 2006.
- Implementation — Good Implementation
  - Kerala (EMS Nambodiriprasha)
  - WB (CPIM)
- Challenges in digitisation

Main Topic 18

Horti/livestock/aquaculture/food processing

- % of GVA
  - 33
  - 30
    - 2014-15 — 24%
    - 2022-23 — 30%
  - 6.7
  - FPI
    - agri — 11.5%
    - manufacturing — 10%
- CAGR (Decadal)
  - 2.82% (Production wise)
  - 7.13%
  - 6.75%
  - 5.35% (2015-23)
- % of Agri exports
  - 7.5
  - 9.3
  - 15
  - 23%
- Employement
  - -
  - 21 Mn
  - 28 Mn
  - 70L / 12% of formal workforce
- Mission
  - Horti
    - Mission for Integrated Development of Horticulture (MIDH) — fruits,vegetables, flowers from prod to post harvest
    - National Horticulture Board (NHB) — post harvest infra in integerated manner
    - horticulture CDP
  - Livestock
    - NDDB — promote farmer owned dairy and cooperatives
    - Rashtriya Gokul Mission — indigenous bovines and enhance milk produciton and producivity
    - National Livestock Mission — sustainable dev of livestocks
  - Aquaculture
    - PM Matsya Sampada Yojna — Blue revolution
      - production, quality
      - sustainable and responsible
    - Fisheries and Aquaculture Infrastructure Development Fund (FIDF) — fish inra — harbours to cold chains
- Concerns
  - Horti — Climate change, water scarcity, post-harvest losses
  - Livestock — Greenhouse gas emissions, resource depletion, low productivity
  - Aquacultre — Environmental impact (mangrove loss), water pollution, disease outbreaks

Main Topic 19

Food Processing

- Intro — We export what we should eat and waste what we shoudl Process
- Conclusion — Backbone of AtmaNirbhar krishi, potential to linking farms into international food chains and reduce rural distress.
- GDP Contri
  - Agriculture — 11.5%
  - Manufacturing — 10%
- CAGR (2015-16 - 2022-23) — MOSPI
  - 5.35%
  - 1.61 -> 1.92 L Cr.
- % of total agri exports
  - 23%
  - Rising CAGR @13%
  - 13 Bn in 2022-23
- % of total organised employement
  - 12%
  - 70L
- Investment growth in last 7 year — 7.25%
- Budget — National Institute of Food Technology, Entrepreneurship and Management to be set up in Bihar
- Post Harvest Losses in Perishables — 20%
- Issues
  - Low Level of Processing
    - India — 10% of agri produce
    - Developed countries — 60-80%
  - Fragmented supply chains — lack of storage — 37 mt
  - lack of processing infra — testing labs, effluent treatment plants, modern processing units, etc.
  - Credit access — 6% of MSME in food processing recive it
  - Lack of Skilled Manpower — 80% unskilled/ informal — NSDC , 2023
  - Red Tapism
- Govt Stance
  - Economic
    - FDI — 100% thtough automatic route
    - PLI
      - in food processing sector (Mozarella Cheese)
      - In
        - Ready to eat
        - Millet
    - PSL
    - Low GST — 0/5%
  - Operation Green
    - Originally for — TOP
    - Now expanded to — (TOTAL) 22 perishable items
    - Aim — Price Stabilisation
    - method — Integerated logistics development
      - support of FPO
      - modernise agri logistics
      - processing facilities
      - managaement
  - PM-Kisan sampada Yojna
    - Cold storage, food parks, etc.
    - Investment — 10,000 Cr
    - Jobs generated — 6L
  - Market
    - ODOP
    - e-NAM
    - Agri Infra Fund
      - 1L Crore
      - Farm gate Infra
      - Primary Processing
- Need to be Done
  - Agri-export zone
  - Cluster Developement — JackFruit Cluster Kerala
  - Credit acess
  - Strengthen FPO and contract farming
  - Skill Development under Skill India Mission
    - Food Safety
    - Packaging

Farmer Debt

- % of household spend more than they earn — 70%
- % in debt due to rise in farm input cost — 50%
- NSSO data

Irrigation

- land under irrigation of Gross Cropped Area — NITI Aayog
  - 2016 — 49%
  - 2021 — 55%
- rice and sugarcane take — 60% of water available for irrigation
- Rainfed Agriculture — % of Prduction
  - Millets — 90%
  - Oil Seeds — 80%
  - Livestock/Cotton — 60%
- %of irrigated area under groundwater irrigation — 60%
- %of ground water usage for irrigation — 90%
- % of global fresh water resource in India — 4%
- Solution
  - Microirrigation can save upto [Ashok Dalwai Committee] — 40% of water
  - Major cause of water inefficiency [NITI Aayog]
    - Electricity subsidy
    - Use graded Metered electricity
- PMKSY
  - Accelerated irrigation benefit Programme
  - per drop mroe crop
  - Har Khet ko Pani — command area developemnt
  - integerated watershed development

Cropping Intensity

- MoAFW
  - 1950 — 111%
  - 2019-20 — 151%
  - Area Under Food Grain (Cross Cropped Area)
    - 1980 — 73%
    - 2022 — 62%
  - Coarse cereals Area
    - 1980 — 22%
    - 2022 — 9%
  - Oil Seed
    - 1980 — 9%
    - 2022 — 15%
  - HortiCulture
    - 2001 — 8.5%
    - 2022 — 14%

Crops and produce

- Cereal
  - 2023 — 332 mt
  - 1947 — 50 mT
  - Expected rise this year
    - 6.6%
    - 354 mt
  - Productivity — FAO
    - Wheat — 3.4-3.6 Ton
    - Rice — 2.8 Ton
    - India — ~3 tonne / hectare
    - China — ~6.1 tonne/ hectare
- Millets
  - Demand per capita per month in rural india
    - 1962 — 4.4Kg
    - 1983 — 3.3 Kg
    - 2011 — 0.4Kg
    - 90% decline from pre green reovlution
  - Export — 2nd Largest in world
  - Productivity — 1.9 tonne per hectare — MoAFW
  - Rise as IYoM capagain — 40% growth in sales — Amazon
  - why millets
    - Climate Resilence — <30 cm rain
    - staple crop for tribals — 60% — SECC
    - Benefits — Health
      - Gluten free ;
      - high nutrients
        - Ragi — 10x calcium than wheat or Rice
        - Bajra — Iron
      - low glycemic index
      - Anti-oxidants
  - Conclusion — Poor Man Grain -> Smart Food
- Horticulture
  - 352 mt — 2023
  - 2nd largest in world Production
  - pick 4 vital stats from above + change in area
  - Why a good sector
    - High income — 5-10x in apple relative to wheat
    - Labour Intensive — planting, harvesting, etc.
    - Nutritional Security — Hidden Hunger
    - Exports — $3.8 Bn
  - Challenges
    - Low Processing
      - 2-3% in fruits — MoSPI
      - Global — 60%
    - Post Harvest Loses — 20%
    - Fragmented Supply Chain
    - Price Volatility
    - Credit Flow — <5% of total agri credit
    - Climate Change Vulnearabilty
  - Govt Stance
    - MIDH (subsumes NHM)
      - production
      - infra
      - market
    - National Horticulture Board
      - Financial assistance for cold storage
      - Nurseris
    - Operation Greens : TOP -> TOTAL (incudes fruits and vegetables) — subsidies on transport and storage
    - Horticulture CDP
  - Way Forward
    - CDP — Kerala CDP Jackfruit
    - FPO promotion and contract farming — Banana in Jalgaon
    - Digitisation — Kisan Sarthi App
      - Pest forecasting
      - AI Advisory
    - Export oriented porduction — Grapes/Litchis
    - Cold Storage — through solar power
    - Crop Insurance
- Pulses
  - Protein by weight — 20-25%
  - Largest
    - Producer — 25%
    - Consumer — 27%
    - Importer — 14%
    - Import dependence — 25%
  - Solution: — Integrated Schemes of Oilseeds, Pulses, Oil palm and Maize or ISOPOM,
- Oil seeds — Import Dependence — 55%

Parlance

- water intensive — water guzzling crop
- dropiing water levels — conversion of indo-gangetic plain into dark zones
- 'bait to plate' tracaibility using IT technology in aquaculture
- human back to machine back via shift from bag packing to containerisation
- fork to farm
- Evergreen revolution coined by MS Swaminathan
  - to address shortcoming of Green Revolution
  - producing more with less land, less water, less pesticides
  - to achieve sustainiable agriculture
- NCF National commssion on Farmers — swaminathan
- DFI — dalwai committee
- high level committe — HLC — shanta kumar

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